Artifact-Aware Learning for Tooth and Pulp Segmentation in CBCT

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Abstract. Metal restorations produce local intensity artifacts in cone-beam computed tomography (CBCT). Pulp chambers and root canals are small and often have weak boundaries. Together with limited voxel-wise annotations, these factors make tooth and pulp segmentation difficult. We present ArtifactAware-SSL, a 3D semi-supervised framework that uses artifact cues for tooth and pulp segmentation. First, each CBCT volume is resampled and normalized, and a binary high-density proxy is added as a second input channel. Then, a residual 3D U-Net is pretrained on all available scans by reconstructing patches degraded by simulated metal-like artifacts, cube masking, and resolution reduction. Finally, the pretrained network initializes an exponential moving average teacher–student model. Labeled patches provide segmentation and reconstruction supervision, while confidence-filtered teacher predictions on unlabeled patches supervise a strongly perturbed student through consistency and pseudo-label losses. At inference, patch-center selection produces whole-volume labels without storing dense 65-class probabilities. The submitted system achieved a composite score of 0.474 and ranked eighth in the STS 2026 challenge.

Keywords: Cone-beam computed tomography · Tooth and pulp segmentation · Semi-supervised learning · Metal artifact · Mean teacher

1 Introduction

Cone-beam computed tomography (CBCT) provides volumetric views of teeth, roots, and surrounding maxillofacial structures for diagnosis and treatment planning [1,9]. Automatic segmentation can reduce the manual work needed to construct patient-specific 3D models, and recent systems can extract several oral structures from a full CBCT scan [7]. Pulp segmentation remains more difficult than segmentation of the tooth body. Pulp chambers and root canals occupy few voxels, extend across many slices, and often have weak contrast against dentine [3,15]. Differences in voxel size and intensity distribution introduce further variation between scans. Metal crowns, fillings, implants, and orthodontic appliances

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can also produce saturated regions, streaks, and shadows that obscure narrow anatomical boundaries.

Dense annotation of a 3D CBCT scan is laborious and requires anatomical expertise. Semi-supervised learning addresses this limitation by combining voxel-wise supervision from a small labeled set with regularization or pseudo-label supervision from a larger unlabeled set. Earlier STS challenges established semi-supervised CBCT benchmarks for tooth segmentation and later extended them to tooth and pulp segmentation and CBCT–IOS registration [14,13]. The STS 2026 setting continues this line of work, with many more unlabeled scans than labeled scans and an emphasis on robust tooth and pulp segmentation.

U-Net, 3D U-Net, and nnU-Net are established backbones for volumetric segmentation [8,2,5]. Mean Teacher stabilizes targets on unlabeled data by averaging model parameters over training [11], and weak-to-strong consistency requires predictions to remain stable under stronger perturbations [16]. Variance-reduction and selective pseudo-labeling methods reduce the effect of unreliable targets and class imbalance in semi-supervised medical image segmentation [17,4]. Disruptive autoencoding instead pretrains 3D representations by reconstructing images degraded by masking, noise, or resolution reduction [12]. In dental CBCT, U-Mamba2-SSL combines disruptive pretraining with input- and feature-level consistency and confidence-filtered pseudo-labels [10]. Efficient full-resolution nnU-Net has also shown that teeth and root canals can be segmented in a single whole-volume pipeline [6]. These methods do not, however, incorporate a spatial cue for high-density material into both pretraining and semi-supervised segmentation. As a result, predictions near severe artifacts may still produce unreliable pseudo-labels.

ArtifactAware-SSL addresses this issue in three stages. Each scan is first resampled and normalized, and a binary high-density proxy is appended as a second input channel. A five-level residual 3D U-Net is then pretrained on all available scans by reconstructing patches degraded by simulated metal-like artifacts, cube masking, and downsampling. The pretrained network initializes an EMA teacher–student model for segmentation. The teacher predicts the original unlabeled view, the student receives stronger image and deep-feature perturbations, and only confident teacher predictions contribute to the consistency and pseudo-label objectives. During whole-volume inference, a center-prioritized sliding window produces the final mask without retaining dense class-wise logits.

2 Method

2.1 Overview

Let $\mathcal{D}_l = \{(x_i, y_i)\}_{i=1}^{N_l}$ and $\mathcal{D}_u = \{u_j\}_{j=1}^{N_u}$ denote the labeled and unlabeled CBCT sets, where x is a raw volume and y is its voxel-wise tooth–pulp annotation. After preprocessing, $\bar{x} \in \mathbb{R}^{D\times H\times W}$ denotes a normalized image patch and m its high-density proxy. The symbols D, H, and W denote the spatial dimensions, and v indexes a voxel. Channel-wise concatenation is written as cat,

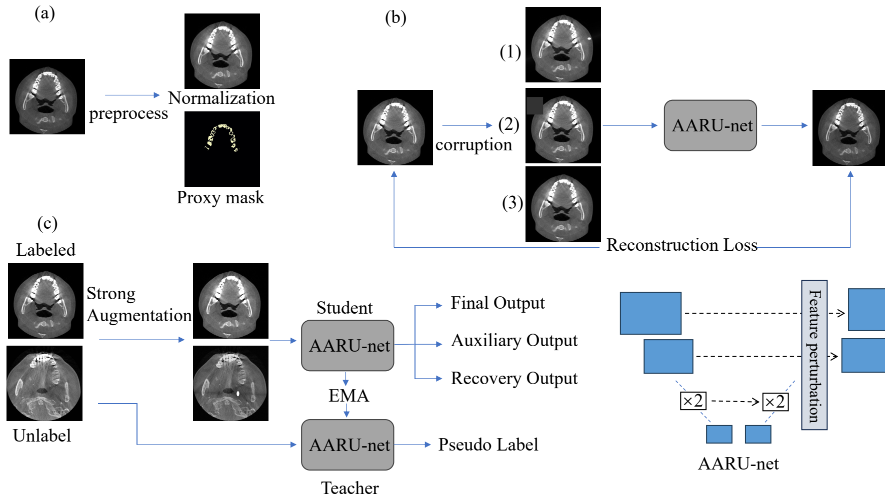

Fig. 1. Overview of ArtifactAware-SSL. (a) Each CBCT scan is resampled and normalized, and a high-density proxy is computed. (b) Metal-like corruption, cube masking, or low-resolution degradation is applied during reconstruction pretraining. (c) During semi-supervised training, labeled patches supervise the main, auxiliary, and reconstruction outputs. For an unlabeled patch, the EMA teacher provides confidence-filtered pseudo-labels, while the student receives input and feature perturbations.

so the two-channel input is $z = \mathrm{cat}(\bar{x}, m)$. The artifact-aware residual U-Net (AARU-Net) and its parameters are denoted by F_θ and θ, respectively.

As shown in Fig. 1, the framework has three stages. Stage I resamples and normalizes each scan and computes the high-density proxy. Stage II trains AARU-Net to reconstruct patches after controlled corruption. Stage III transfers the pretrained parameters to an EMA teacher–student model. Labeled patches supervise the main segmentation output, two auxiliary outputs, and the reconstruction output. Unlabeled patches contribute confidence-filtered consistency and pseudo-label losses to the student's main output.

2.2 Stage I: Preprocessing

All CBCT scans were resampled to approximately 0.3-mm isotropic spacing, using cubic interpolation for images and nearest-neighbor interpolation for labels. Intensities were clipped at the case-wise 0.5th and 99.5th percentiles and then standardized by the case-wise mean and standard deviation. Resampling reduces variation in physical scale, while clipping limits the influence of extreme values. Voxels at or above the 99.7th percentile formed a binary high-density proxy, which was concatenated with the normalized image. This proxy indicates regions that may contain dense materials associated with strong artifacts. It is not a metal annotation and can also include enamel or other dense tissue.

2.3 Stage II: Corruption-Based Pretraining

Stage II trains F_θ on all scans in $\mathcal{D}_l \cup \mathcal{D}_u$ while withholding their segmentation labels. The task is to reconstruct an image patch after artificial corruption, with the input and target kept on the same spatial grid. It therefore uses no voxel-wise annotation and encourages the network to learn 3D dental structure from both labeled and unlabeled scans [12].

Three corruption families are sampled during pretraining. Metal-like corruption adds a high-density source and directional bright or dark streaks to approximate the image-domain appearance of artifacts near restored teeth. Cube masking removes a contiguous 3D region so that its content must be inferred from nearby voxels and slices. Low-resolution degradation downsamples a patch and returns it to the original grid, weakening narrow pulp regions and sharp boundaries. In each case, reconstruction requires the network to recover local structure from the surrounding anatomical context.

Let ξ identify a sampled corruption and let $\mathcal{C}_\xi$ denote the corresponding operator. It produces a corrupted patch $\tilde{x}$ and an affected region map a

$$(\tilde{x}, a) = \mathcal{C}_\xi(\bar{x}), \qquad a(v) \in [0, 1]. \tag{1}$$

The map a records the support and relative strength of the corruption. We combine it with the high-density proxy to obtain

$$m_a(v) = \max\{m(v), a(v)\}, \qquad \tilde{z} = \mathrm{cat}(\tilde{x}, m_a), \tag{2}$$

where m_a is the joint emphasis map and $\tilde{z}$ is the corrupted two-channel input.

Let R_θ denote the reconstruction branch. Its output $\hat{x}$ is a bounded residual correction to $\tilde{x}$

$$\hat{x} = \tilde{x} + \gamma \tanh(R_\theta(\tilde{z})), \tag{3}$$

where $\gamma > 0$ limits the correction magnitude.

The reconstruction loss gives additional weight to voxels in the high-density proxy and the artificially corrupted region. With $w(v) = 1 + \lambda_a m_a(v)$, where $\lambda_a \geq 0$ controls the additional weight, the pretraining objective is

$$\mathcal{L}_{\mathrm{pre}} = \frac{\sum_v w(v)\ \mathrm{SmoothL1}(\hat{x}(v), \bar{x}(v))}{\sum_v w(v)}. \tag{4}$$

The target is the original observed patch, not an artifact-free reference. Pretraining therefore learns recovery from controlled corruption rather than supervised metal-artifact removal. The resulting parameters initialize Stage III.

2.4 Stage III: Semi-Supervised Segmentation

The student F_{θ_s} is initialized from Stage II, and the teacher F_{θ_t} starts from the same AARU-Net parameters.

For a labeled pair (x, y), the student receives an augmented two-channel view and produces one full-resolution segmentation output, two lower-resolution auxiliary outputs, and a reconstruction output. The loss for the final output is $\mathcal{L}_{\text{seg}} = \mathcal{L}_{\text{Dice}} + \mathcal{L}_{\text{CE}}$. Including the auxiliary segmentation losses and the reconstruction loss $\mathcal{L}_{\text{rec}}$, the labeled objective is

$$\mathcal{L}_l = \mathcal{L}_{\text{seg}} + \sum_{k=1}^{2} \lambda_k \mathcal{L}_{\text{seg}}^{(k)} + \lambda_r \mathcal{L}_{\text{rec}}, \tag{5}$$

where λ_k and λ_r weight the auxiliary losses and reconstruction loss, respectively. The two auxiliary heads branch from intermediate decoder scales and use nearest-neighbor downsampled labels with the same segmentation loss. They provide direct supervision to shared intermediate features but are not used for unlabeled training or inference. The reconstruction output follows Eq. (3), and its loss is computed as in Eq. (4) on the augmented labeled view.

For an unlabeled patch, the teacher receives the normalized image and its high-density proxy, whereas the student receives a strongly perturbed view of the same patch. This weak-to-strong scheme [16] combines CBCT-specific input corruption with feature perturbation [10]. Feature perturbation is applied only to the two deepest student encoder features and combines channel dropout, suppression of strongly activated spatial regions, and multiplicative noise. The perturbation discourages reliance on a small set of dominant features. Only the final segmentation output contributes to the unlabeled objectives; the auxiliary and reconstruction outputs are excluded.

Let $p(v)$ and $q(v)$ be the student and teacher class-probability vectors at voxel v, respectively. The KL term aligns these distributions, while the most likely teacher class provides the hard pseudo-label.

Let $\mathcal{C}$ be the class set and τ the confidence threshold. The confidence mask M_τ and retained pseudo-label $\tilde{y}$ are defined as

$$M_\tau(v) = \begin{cases} 1, & \max_{c \in \mathcal{C}} q_c(v) \geq \tau, \\ 0, & \text{otherwise}, \end{cases} \qquad \tilde{y}(v) = \arg\max_{c \in \mathcal{C}} q_c(v). \tag{6}$$

The consistency term matches the teacher and student probability distributions, whereas the pseudo-label term trains the student toward the class selected by the teacher. Let $Z_\tau = \max\{1, \sum_v M_\tau(v)\}$ denote the number of selected voxels, lower-bounded by one. The two losses are

$$\mathcal{L}_{\text{con}} = \frac{1}{Z_\tau} \sum_v M_\tau(v) D_{\text{KL}}(q(v) \| p(v)), \qquad \mathcal{L}_{\text{pl}} = -\frac{1}{Z_\tau} \sum_v M_\tau(v) \log p_{\tilde{y}(v)}(v). \tag{7}$$

The masked sums are zero when no voxel satisfies the threshold. Confidence filtering thus excludes uncertain teacher predictions from direct student supervision [17,4].

Only the student receives gradients. After each student update, the teacher parameters are updated by EMA [11]:

$$\theta_t \leftarrow \alpha \theta_t + (1 - \alpha) \theta_s, \tag{8}$$

where, α is the EMA decay. The complete Stage III objective is

$$\mathcal{L} = \mathcal{L}_l + r(t)\left(\lambda_{\text{con}}\mathcal{L}_{\text{con}} + \lambda_{\text{pl}}\mathcal{L}_{\text{pl}}\right), \tag{9}$$

where λ_{con} and λ_{pl} weight the two unlabeled losses, and $r(t)$ gradually activates them over training progress t. This delay allows the teacher predictions to stabilize before they begin to supervise the student.

3 Experiments

3.1 Dataset

The dataset contains 30 labeled and 299 unlabeled CBCT volumes for training, together with 20 validation CBCT volumes whose annotations are withheld.

3.2 Implementation Details

The method was implemented in PyTorch with automatic mixed-precision training. Structure-restoration pretraining was followed by semi-supervised segmentation with an EMA teacher. The complete training configuration is summarized in Table 1.

Table 1. Training configuration.

Setting	Value
Network	Five-level residual 3D U-Net; encoder widths $(24, 48, 96, 160, 224)$; residual-block counts $(1, 2, 2, 2, 2)$; Group Normalization; Leaky ReLU
Outputs	65-class main head; auxiliary-head weights 0.25 and 0.125; $\gamma = 0.20$, $\lambda_a = 3$, and $\lambda_r = 0.15$
Patch sampling	Patch size $96\times192\times192$; batch size 1; foreground-centered sampling probability 0.67
Pretraining	8,000 steps; corruption-family probability 0.35; at most two metal-like sources
Feature perturbation	Two deepest encoder levels; probability 0.5; channel dropout, response suppression, and multiplicative noise
EMA teacher and SSL	$\alpha = 0.995$; $\tau = 0.75$; $\lambda_{\text{con}} = 1$; $\lambda_{\text{pl}} = 0.15$
Unlabeled-loss schedule	Disabled for the first 15% of training and ramped up over the next 20%
Optimization	AdamW; initial learning rate 2×10^{-4}; weight decay 10^{-5}; gradient clipping 12; cosine decay to 10^{-5}
Segmentation and selection	50,000 steps; evaluation every 1,000 steps on 48 fixed pseudorandom validation patches; highest foreground macro Dice

For whole-volume inference, the selected EMA model used the same patch size as in Table 1 with 25% overlap. We applied neither test-time augmentation nor connected-component removal. Instead of retaining dense 65-channel whole-volume logits, the implementation assigned each voxel the label predicted by the patch with the nearest center. This rule reduced memory use. The final mask was mapped to the original label IDs and image geometry.

4 Results and Discussion

4.1 Challenge Result

In the STS 2026 challenge evaluation, ArtifactAware-SSL achieved a composite score of 0.474 and ranked eighth on the leaderboard. Figure 2 provides a qualitative comparison of two cases from the internal validation set. The highest-scoring case shows close agreement between the prediction and reference annotation in both the 3D tooth arrangement and the representative axial slice. By contrast, the lowest-scoring case shows more pronounced discrepancies in the extent and class assignment of the segmented structures.

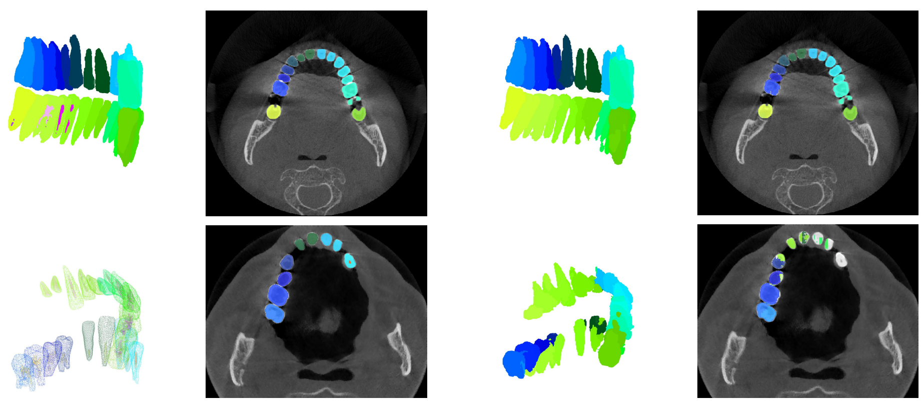

Fig. 2. Qualitative results of ArtifactAware-SSL on the internal validation set. Ground-truth annotations (a) and model predictions (b) are shown as a 3D rendering and a representative axial slice. The top row corresponds to the highest-scoring case, and the bottom row corresponds to the lowest-scoring case.

4.2 Discussion and Limitations

The design reflects two practical constraints. Artifact handling must work without paired clean CBCT scans or projection data, so the high-density proxy and bounded reconstruction branch condition the network without replacing the observed anatomy. In addition, storing a whole-volume probability map for 65 classes is expensive. Patch-center selection lowers output memory, but discards the class probabilities that could otherwise be blended across overlapping windows. These choices simplify the submission pipeline, although the eighth-place result shows that its robustness and fine-structure accuracy still need improvement.

Several limitations follow directly from the design. The 99.7th-percentile proxy is nonspecific and can mark enamel while missing streaks far from their source. Image-domain ellipsoids and stripes do not reproduce the physics of projection-domain metal artifacts. Fixed confidence thresholding can accept a high-confidence but incorrect teacher prediction, especially for rare pulp labels, and patch-center selection discards probability information at window boundaries. The current study also lacks controlled ablations, per-class organizer metrics, and evaluation stratified by artifact severity. Further experiments should test uncertainty-aware thresholds, pulp-aware sampling or topology constraints, and calibrated probability fusion in place of hard stitching. Dice, surface, identification, runtime, and memory measurements are also needed to determine which part of the pipeline accounts for the gap to higher-ranked submissions.

5 Conclusion

ArtifactAware-SSL combines a high-density proxy, corruption-based pretraining, and EMA teacher–student learning for semi-supervised tooth and pulp segmentation. The method uses both labeled and unlabeled CBCT scans without requiring paired artifact-free images. Patch-center inference further limits memory use during whole-volume prediction. The submitted system achieved a composite score of 0.474 and ranked eighth in the STS 2026 challenge. Improving the treatment of small pulp structures and uncertainty near metal artifacts remains an important direction for further work.

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Table 2. Checklist Table. Please fill out this checklist table in the answer column.

Requirements	Answer
A meaningful title	Yes
The number of authors (≤ 6)	2
Author affiliations, Email, and ORCID	Yes
Corresponding author is marked	Yes
Validation scores are presented in the abstract	Yes
Introduction includes at least three parts: background, related work, and motivation	Yes
A pipeline/network figure is provided	Fig. 1
Pre-processing	Page 3
Strategies to use the partial label	Page 4
Strategies to use the unlabeled images	Pages 4–5
Strategies to improve model inference	Page 6
Post-processing	Page 6
Dataset and evaluation metric section is presented	Page 7
Environment setting table is provided	Page 6
Training protocol table is provided	Table 1
Ablation study	No
Efficiency evaluation results are provided	Page 7
Visualized segmentation example is provided	Page 7
Limitation and future work are presented	Yes
Reference format is consistent	Yes

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